Solved current papers --- February 2019

Zoo507-Principle of Animal Ecology

Q.bio degradable and non-degradable pollutants.

There are two main types of pollutants.

• Biodegradable pollutants

• Non-degradable pollutants

Biodegradable pollutants are harmful materials that can be broken down by natural processes. Examples are human sewage and newspapers.

Non-degradable pollutants are harmful materials that natural processes cannot break down. Examples are toxic chemical elements such as lead, mercury, and arsenic.

Q.Industrial Ecology ?

• The design of the industrial infrastructure such that it consists of a series of interlocking "technological ecosystems" interfacing with global natural ecosystems.

• Industrial ecology takes the pattern and processes of natural ecosystems as a design for sustainability. It represents a shift in paradigm from conquering nature to becoming nature.

Q.allogenic and autogenic succession?

Autogenic succession

▪ In autogenic succession the successional process

▪ is controlled by living or biotic component of an ecosystem

Allogenic succession

▪ The entire successional process is controlled by

▪ non-living or abiotic components of ecosystem.

▪ Ecological succession is a process of gradual changes that causes the plants, animals, fungi and microorganism to become established in an area.

Q. El-nino effect ???

Another type of short-term climate change is the so-called El Nino effect, which occurs every 2 to 7 years around Christmas time, when the typical weather pattern in the Pacific Ocean breaks down.

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The trade winds weaken, and low pressure establishes itself in the central Pacific. It causes the winds to blow into the Pacific Basin from the west.

This change causes warm surface water to move toward South and Central America, reversing the directions of some currents and placing warm water along the coasts.

The warm water produces heavy rain over the coastal deserts, subjecting them to disastrous flooding and erosion. The western United States also experiences major flooding events as a result of El Nino.

Q. Overexploitation????

Overexploitation of resources (or using resources at an unsustainable rate) is a critical problem in conservation.

The overexploitation or nonsustainable use of wildlife is closely linked and plays an increasing role in the loss of biodiversity

Overharvesting, nonsustainable use, and the illegal trade in some species are threatening not only their continued survival but also that of ecosystems and the livelihoods of communities and local economics that depend upon them.

Unsustainable hunting, fishing, logging, or gathering of wild populations leads to their commercial, ecological or global extinction.

Q. factors responsible for genetic diversity loss in small populations ????

1. Genetic drift,
2. Founder effect,
3. Demographic bottleneck,
4. Inbreeding

• Genetic drift

• " Random change in gene frequency within a population" (Meffe et al.),

• Not necessarily adaptive, Stronger in small populations

• Founder effect,

• "The establishment of a new population by a few original founders which carry only a small fraction of the total genetic variation of the parental population." (Ernst Mayr), Alleles may be lost, Differentiation from the parental population, Example: Irish Potato Famine

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• Demographic bottleneck,

• Population suffers reduction in size and then recovers, Random losses of genetic diversity,

• Usually associated with catastrophic events or diseases,

• Genetic variation doesn't rebound from a decrease as quickly as population size.

• Inbreeding

• Inbreeding is breeding between close relatives,

• Leads to a reduction in genetic diversity (inbreeding depression: reduced health and fitness),

Q. Two types of energy on the base of energy

Quality, ????

1. High quality energy

2. Low quality energy

High quality energy

High-quality energy is concentrated form of

energy and has a high capacity to do useful work.

Examples are,

▪ Very high-temperature heat

▪ Nuclear fission

▪ Concentrated sunlight

▪ High-velocity wind

▪ Energy released by burning natural gas, gasoline, or coal.

▪ Low quality energy

▪ A dispersed form of energy that has a little capacity to do useful work.

▪ Heat energy of moving water molecules is of low quality because it is dispersed in water

molecules

▪ that lowers its temperature.

▪ The total amount of heat stored in the Atlantic Ocean is greater than the amount of high

quality chemical energy stored in all the oil deposits of Saudi Arabia. But wide dispersion of this ocean's heat energy makes it unable to do useful work.

Q. difference bw urban and landscape ecology????

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Urban ecology

• For ecologists, urban ecology is the study of ecology in urban areas, specifically the relationships, interactions, types and numbers of species found in urban habitats.

• Also, the design of sustainable cities, urban design programs that incorporate political, infrastructure and economic considerations.

Landscape Ecology

• "Landscape ecology is concerned with spatial patterns in the landscape and how they develop, with an emphasis on the role of disturbance, including human impacts" (Smith and Smith).

• It is a relatively new branch of ecology, that employs Global Information Systems.

• The goal is to predict the responses of different organisms to changes in landscape, to ultimately facilitate ecosystem management.

Q. What is difference between hibernation winter sleep????

-Hibernation

A condition in which animals become inactive usually in cold environments for their survival.

-Hibernation vs winter sleep

True hibernation occurs in small animals in which temperature substantially falls to remain alive in cold environment. Animals remain inactive for weeks or even for months.

Hibernation in large animals like bears is termed as winter sleep.

Body temperature does not falls substantially and sleeping animals can wake and active quickly. It lasts for short period of time.

Large animals have more energy reserves that enables them to survive in winter.

Q.Name two types of symbiosis.????

Common form of symbiosis are

o Commensalism

o Mutualism

What is greenhouse effect?????

Greenhouse effect

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Carbon dioxide released from burning fuels accumulates in the atmosphere and prevent the escape of heat coming from earth. This effect is termed as Greenhouse effect.

Q. Aposematic coloration with two examples?????

-Aposematic coloration

Some animals protect themselves by being dangerous or distasteful to predators by using prominent coloration. These color patterns are known as warning or aposematic coloration.

The sharply contrasting white stripes of a skunk and bright colors of poisonous snakes give similar messages

They flash a warning: “Eating me is risky.” Examples are brilliantly colored poisonous frogs is another example.

Q. Competition difference between interspecific and interspecific competition

Interspecific competition?????

All organisms require space and food for their survival. Requirement of both space and food causes competition between the species as well as among the same species.

The competition between different species for purpose of space and food is called Interspecific competition.

While competition within the same species is known as Intraspecific competition.

When members of different species compete for resources, one species may be forced to move or become extinct, or the two species may share the resource and coexist.

Most studies have shown that competing species can coexist.

Coexistence can occur when species utilize resources in slightly different ways.

Q. Define biodiversity?????

Biodiversity is a vital renewable resource.

Biological diversity, or biodiversity, is the variety of all life forms on the earth.

The variety of life on Earth at all its levels, from genes to ecosystems, and the ecological and evolutionary processes that sustain it.

Q.difference between direct and indirect exploitation?????

o Direct exploitation

o Indirect exploitation

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Direct exploitation:

Direct exploitation ranges from commercial activities such as logging operations or trade in endangered species to subsistence hunting.

Indirect exploitation:

Indirect exploitation encompasses the unintentional mortality of non-target species such as fish or turtles killed as by-catch in fishery operations.

Both endanger species around the world.

Define bio magnification?????

A phenomenon called bio magnification is well known for mercury and DDT.

A few toxic substances, often present in industrial waste waters, can undergo biological magnification (Bio magnification) in the aquatic food chain.

Bio magnification refers to increase in concentration of the toxicant at successive trophic levels.

Q phosphorus cycle?????

-Phosphorus cycle

Phosphorus circulates through water, the earth's crust, and living organisms in the phosphorus cycle.

In contrast to the cycles of water, carbon, and nitrogen, the phosphorus cycle does not include the atmosphere. The major reservoir for phosphorous is phosphate salts containing phosphate ions (PO_4^{-3}) in terrestrial rock formations and ocean bottom sediments. The phosphorus cycle is slow compared to the water, carbon, and nitrogen cycles.

Phosphorous is a component of biologically important molecules such as,

1. Nucleic acids
2. Energy transfer molecules (ADP and ATP)
3. Vertebrate bones and teeth.

Q. how domestic waste is collected and disposed?????

Collection and Disposal:

Domestic wastes are collected in small bins, which are then transferred to community bins by private or municipal workers. From these community bins, these are collected

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and carried to the disposable site.

At the site, garbage is sorted out and separated into biodegradable and nonbiodegradable materials. On-biodegradable materials such as plastic, glass, metal scraps etc. are sent for recycling. Biodegradable wastes are deposited in landfills and are converted into compost.

The waste if not collected in garbage bins, finds its way into the sewers. Some of it is eaten by cattle. Non-biodegradable wastes like polythene bag, metal scraps, etc. if swallowed by cattle can cost their lives.

All domestic wastes should be properly collected and disposed. The poor management causes health problems leading to epidemics due to contamination of ground water. It is especially hazardous for those who are in direct contact with the waste.

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